Table 1 Drop-out analysis comparing baseline characteristics of Drop-Outs and Participants at T3

	Drop-outs ($n = 316$)		Participants T3 ($n = 231$)		Test statistic (χ^2 or t-test)[f]			
	n (%)	M (SD)	n (%)	M (SD)	χ^2	t	df	*p*
Female gender	168 (53)		133 (58)		0.88		1	.349
Highest education[a]					3.05		3	.385
- No secondary school	0 (0)		0 (0)					
- Lower secondary school[b]	12 (10)		19 (8)					
- Intermediate secondary school[c]	35 (30)		56 (24)					
- Grammar school[d]	25 (21)		45 (20)					
- University[e]	45 (39)		111 (48)					
Age		45.89 (11.57)		45.97 (10.29)		−0.09	542.4	.928
PHQ (sum-score)		13.35 (5.03)		12.30 (5.20)		2.38	486.5	**.018**
Quantitative Demands		54.40 (20.93)		56.40 (22.07)		−1.07	480.24	.285
Emotional Demands		48.44 (32.72)		49.03 (32.41)		−0.21	498.29	.836
Dissolution		33.40 (27.13)		34.31 (26.53)		−0.90	501.59	.697
Decision authority		38.37 (23.72)		42.64 (21.96)		−2.17	515.73	**.030**
Development Opportunities		60.81 (21.63)		66.77 (20.93)		−3.25	504.96	**.001**
Social Support		58.16 (23.16)		60.62 (22.06)		−1.26	508.32	.208

Abbreviations: *df* Degrees of freedom, *M* Mean, *n* Number, *p p*-value, *PHQ* Patient Health Questionnaire, *SD* Standard deviation, *T3* Third follow-up

Statistical significance at $p <= .05$ shown in bold

[a]Education was surveyed at T2. Drop-outs for this variable thus represent individuals participating at T2 but not T3

[b]school-leaving certificate after nine years of school (German: Hauptschule)

[c]school-leaving certificate after ten years of school (German: Realschule)

[d]school-leaving certificate after 12–13 years of school (German: Gymnasium)

[e]General and applied-science university

[f]compares baseline characteristics between Drop-Outs and Participants T3

associated with perceived quantitative demands or perceived job resources. These results underline previous evidence that work-focused CBT can lead to a decrease in job stress [30, 32, 33] and suggests that those effects are enduring over a period of two years. However, these findings should be interpreted cautiously and require confirmation in future studies due to the exploratory study design and the risk that significant findings were only found due to multiple testing. The risk that significant findings were only found due to multiple testing is especially high for perceived dissolution as shown by the insignificant Benjamini–Hochberg adjusted p-value. The finding for perceived emotional demands appears to be slightly more robust, as the Benjamini–Hochberg adjusted p-value remained close to statistical significance and the effect remained statistically significant in the sensitivity analysis excluding participants who reported that they did not work at T3. Our findings further support results of a previous study on an early intervention comprising problem solving and CBT for employees being at risk of sick leave due to depressive disorders, which also found no differences between the IG and CG regarding perception of quantitative job demands, decision authority and social support at follow-up [31].

Regarding practical significance of the findings, the adjusted mean difference of 6.61 points between the IG and CG for perception of emotional demands corresponds to approximately one quarter of a response category on the original five-point scale (25 points per response category) of the COPSOQ. Regarding perceived dissolution, the adjusted mean difference of 5.05 points between the IG and CG corresponds to approximately one fifth of a response category. Even though those differences seem small, one should keep in mind that the intervention was tailored to individual needs. Participants did not uniformly receive components potentially addressing emotional demands and dissolution between work and private-life. Therefore, the observed overall association may underestimate the potential impact of specific intervention components in participants for whom they are indicated. To the best of our knowledge, no established threshold values for practically or clinically meaningful changes exist for COPSOQ scales. However, the strength of effects within our study are comparable to those of mindfulness-based interventions. For example, comparable reductions in general perceived job demands and dissolution were found in a sample of healthy managers after participating in a mindfulness-based intervention [57].

The observed reductions in the perception of emotional demands and dissolution may reflect a combination of cognitive change and behavioral mechanisms rather than actual changes in the work environment. Because potential mechanisms of change were not examined by this study, they should be regarded as hypotheses for future research. It is conceivable that psychoeducation

Table 2 Descriptive analyses of perceived working conditions at T0 and T3

	Intervention group						Control group					
	T0	T3	Paired t-test[a]				T0	T3	Paired t-test[b]			
	M (SD)	M(SD)	n	t	df	*p*	M (SD)	M(SD)	n	t	df	*p*
Quantitative demands	56.90 (21.31)	49.45 (20.69)	137	4.57	136	**<.001**	55.68 (23.23)	51.88 (21.67)	94	2.26	93	**.026**
Emotional demands	50.36 (31.06)	45.62 (31.01)	137	2.40	136	**.018**	47.07 (34.36)	49.34 (29.18)	94	−1.11	93	.272
Dissolution	35.49 (26.96)	26.55 (23.16)	137	4.82	136	**<.001**	32.58 (25.94)	30.19 (22.41)	94	1.047	93	.298
Decision authority	42.88 (22.63)	45.89 (21.28)	137	−1.83	136	.070	42.29 (21.07)	45.48 (22.14)	94	−1.529	93	.130
Development opportunities	66.18 (20.38)	65.81 (18.32)	136	0.12	135	.907	67.64 (21.79)	68.93 (21.56)	94	−0.707	93	.481
Social support	58.29 (21.50)	63.29 (21.19)	135	−3.01	134	**.003**	64.03 (22.53)	66.80 (22.95)	94	−1.185	93	.239

[a]compares perceived working conditions at T0 and T3 within the intervention group

[b]compares perceived working conditions at T0 and T3 within the control group

Abbreviations: *df* Degrees of freedom, *M* Mean, *n* Number, *p p*-value, *SD* Standard deviation, *T0* Baseline, *T3* Third follow-up

Statistical significance at $p <= .05$ shown in bold

Table 3 Multiple linear regression models predicting perceived job demands at T3, without imputation

	Quantitative demands (*n* = 231)				Emotional demands (*n* = 231)				Dissolution (*n* = 231)			
	b	ß	SE	*p*	b	ß	SE	*p*	b	ß	SE	*p*
Outcome variable at T0	0.58	0.61	0.05	**<.001**	0.68	0.72	0.04	**<.001**	0.52	0.61	0.05	**<.001**
Age	0.06	0.03	0.11	.592	0.31	0.11	0.13	**.021**	0.19	0.08	0.13	.147
Gender (ref. male)	4.71	0.11	2.27	**.039**	3.93	0.06	2.64	.138	0.84	0.02	2.76	.760
Education	−0.46	−0.02	1.30	.723	3.33	0.11	1.40	**.018**	0.42	0.02	1.55	.788
Occupation (ref. blue collar)	4.92	0.09	3.24	0.130	−1.84	−0.02	4.02	0.65	−0.68	−0.01	3.50	0.846
PHQ at T0 (ref. sum-score < 11)	−0.85	−0.02	2.24	.706	−1.41	−0.02	2.89	.627	1.48	0.03	2.52	.557
Days of sickness absence at T0 (ref. < 21 days)	−3.55	−0.08	2.43	0.145	−3.76	−0.06	3.17	0.236	−3.30	−0.07	2.87	0.252
Group (ref. IG)	3.36	0.08	2.15	.120	6.61	0.11	2.52	**.009**	5.05	0.11	2.52	**.047**
Model fit	Adj. $R^2 = 0.46$, $F(12,218) = 15.38$, $p < .001$				Adj. $R^2 = 0.59$, $F(12,218) = 28.51$, $p < .001$				Adj. $R^2 = 0.363$, $F(12,218) = 11.91$, $p < .001$			

All model were adjusted for study centre

Statistical significance at $p <= .05$ shown in bold

Abbreviations: *adj.* Adjusted, *b* Not standardized regression estimate, *ß* Standardized regression estimate, *IG* Intervention group, *n* Number, *p p*-value, *PHQ* Patient Health Questionnaire, *ref.* Reference group (scored as 0), *SE* Standard Error, *T0* Baseline, T3 Third follow-up

Table 4 Multiple linear regression models predicting perceived job resources at T3, without imputation

	Decision authority (*n* = 231)				Development opportunities (*n* = 230)				Social support (*n* = 229)			
	b	ß	St.E	*p*	b	ß	St.E	*p*	b	ß	St.E	*p*
Outcome variable at T0	0.56	0.57	0.06	**<.001**	0.54	0.58	0.05	**<.001**	0.48	0.48	0.06	**<.001**
Age	−0.23	−0.11	0.12	.065	−0.13	−0.07	0.11	.263	−0.24	−0.11	0.13	.072
Gender (ref. male)	−0.89	−0.02	2.42	.713	0.79	0.02	2.20	.721	3.91	0.09	2.70	.150
Education	1.06	0.05	1.39	.445	−0.71	−0.10	1.28	.582	0.80	0.04	1.66	.631
Occupation (ref. blue collar)	1.04	0.02	3.44	.763	4.53	0.09	3.10	.146	−6.97	−0.13	3.72	.062
PHQ at T0 (ref. sum-score < 11)	−2.98	−0.07	2.41	.218	−4.06	−0.10	2.21	.067	−1.32	−0.03	2.72	.628
Days of sickness absence at T0 (ref. < 21 days)	1.03	0.02	2.62	.694	−3.69	−0.09	2.36	.120	−3.95	−0.08	3.07	.199
Group (ref. IG)	0.11	0.00	2.32	.962	2.02	0.05	2.10	.339	0.41	0.00	2.67	.878
Model fit	Adj. $R^2 = 0.37$, $F(12,218) = 12.14$, $p < .001$				Adj. $R^2 = 0.38$, $F(12,217) = 12.64$, $p < .001$				Adj. $R^2 = 0.29$, $F(12,216) = 8.83$, $p < .001$			

All model were adjusted for study centre

Statistical significance at $p <= .05$ shown in bold

Abbreviations: adj. Adjusted, *b* Not standardized regression estimate, *ß* Standardized regression estimate, *IG* Intervention group, *n* Number, *p* p-value, *PHQ* Patient Health Questionnaire, *ref.* Reference group (scored as 0), *SE* Standard Error, *ref.* Reference group, *T0* Baseline, *T3* Third follow-up

and development of an individual strain model [10, 14] increased participants' awareness of emotional job demands and blurring boundaries between work and private life, while cognitive restructuring changed participants' perspective how to interpret and evaluate work-related stressors [16–18]. The intervention might have further improved emotional regulation, setting boundaries, and reattributing responsibility for interpersonal demands at work. Furthermore, the intervention explicitly targets increased work-related self-efficacy [23], which may have boosted participants' confidence in managing emotionally demanding situations.

Likewise, behavioral strategies such as problem solving, relaxation and stress coping techniques [10, 14, 19] may have helped participants to better cope with emotionally demanding situations, eventually leading to the feeling that emotional job demands have decreased. The intervention may also have encouraged patients to establish clearer boundaries between work and their private lives. This assumption is supported by qualitative findings showing that employees on RTW after CMD-related sick leave placed greater meaning to their private life and reduced dissolution of work and private life [58].

Last but not least, the intervention-related reduction of CMD symptoms [9] may also have contributed to a more positive perception of emotional job demands and dissolution between work and private life. Depression and anxiety are associated with impaired emotion regulation and cognitive biases in the processing of negative information [26, 27]. Consequently, reductions in these symptoms may result in a more favorable appraisal of working conditions. As described as a downward-spiral process of negative emotion, perceived stress may further increase symptoms of depression and anxiety, which again reinforce the perception of stress [59]. Such reciprocal relationships have also specifically been described for job strain and depressive symptoms [29, 60]. However, the present study cannot disentangle whether improvements in mental health changed participants' perceptions of their job demands, whether changes in perceived job demands contributed to symptom improvement, or whether both outcomes resulted from shared intervention mechanisms.

In contrast, no associations were observed between the intervention and perceived job resources. One possible explanation is that the intervention primarily targeted work-related problems rather than building resources at work. In addition, job resources might be more strongly tied to the organization and a closer contact to the employer or its' occupational health professionals might be needed during the intervention to facilitate adaptations of working conditions. Furthermore, only a minority of participants received the RTW module that encouraged communication with employers about